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Orchestrator

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Jacobi Co-Simulation: One Macro Step $t^\#$ $t^{\#\#}$

Phase 1: Set All Inputs

All inputs frozen at $t^\#$
using outputs from previous step

set $u^\#(t^\#)$ from $y(t^{\#\#})$

inputs set

set $u^\#(t^\#)$ from $y(t^{\#\#})$

inputs set

set $u^\#(t^\#)$ from $y(t^{\#\#})$

inputs set

Phase 2: Parallel Integration

All SUs integrate
independently from $t^\#$ to $t^{\#\#}$

par

[Parallel Execution]

doStep($t^\#, H$)

$x^{\#\#\#} = \#\#(t^\#, x^\#, u^\#)$

$y^{\#\#\#} = \#\#(t^{\#\#}, x^{\#\#\#})$

outputs $y^\#(t^{\#\#})$

doStep($t^\#, H$)

$x^{\#\#\#} = \#\#(t^\#, x^\#, u^\#)$

$y^{\#\#\#} = \#\#(t^{\#\#}, x^{\#\#\#})$

outputs $y^\#(t^{\#\#})$

doStep($t^\#, H$)

$x^{\#\#\#} = \#\#(t^\#, x^\#, u^\#)$

$y^{\#\#\#} = \#\#(t^{\#\#}, x^{\#\#\#})$

outputs $y^\#(t^{\#\#})$

Exchange outputs at $t^{\#\#}$

New outputs available
for next macro step

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